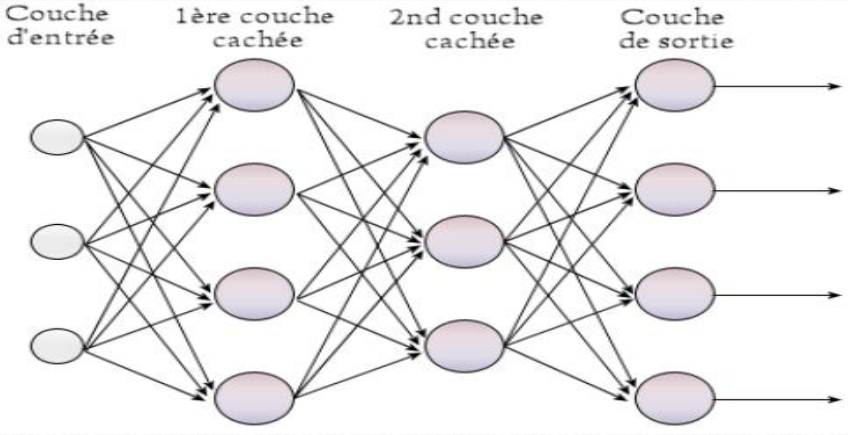
Introduction

- La première version du perceptron était **mono-couche**
- Non capable de résoudre des problèmes non linéaires

→ Perceptron **multicouche**: Ajouter des couches cachées pour résoudre les problèmes non linéaires

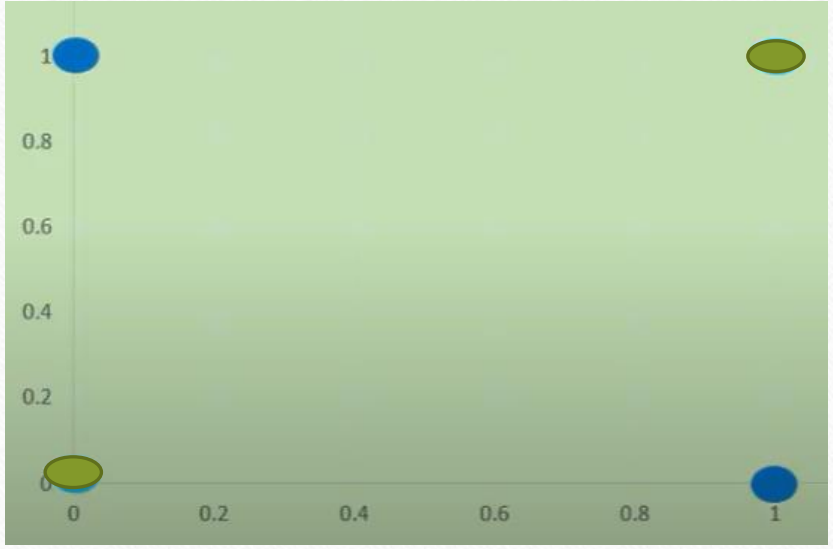


Exemple de classification: problème logique XOR

- Résolution d'un problème logique de type XOR

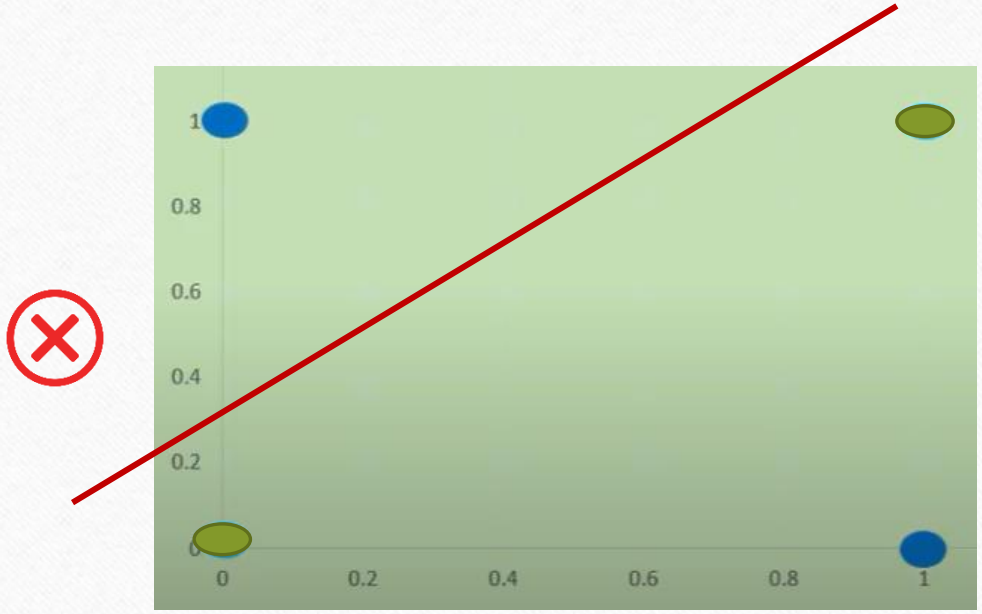
A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0

Exemple de classification: Résolution non linéaire



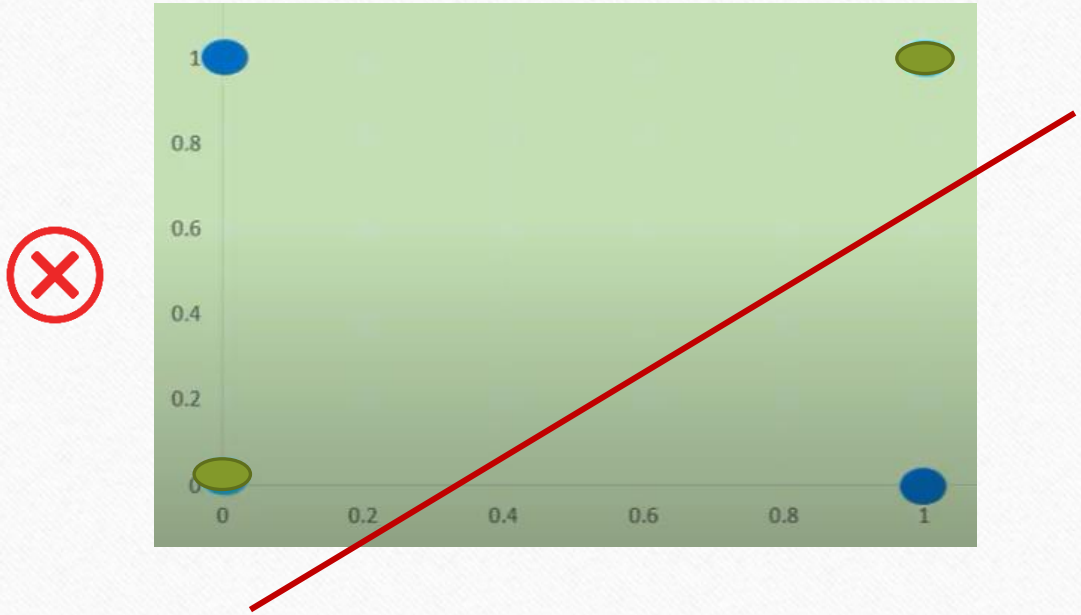
A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0

Exemple de classification: Résolution non linéaire



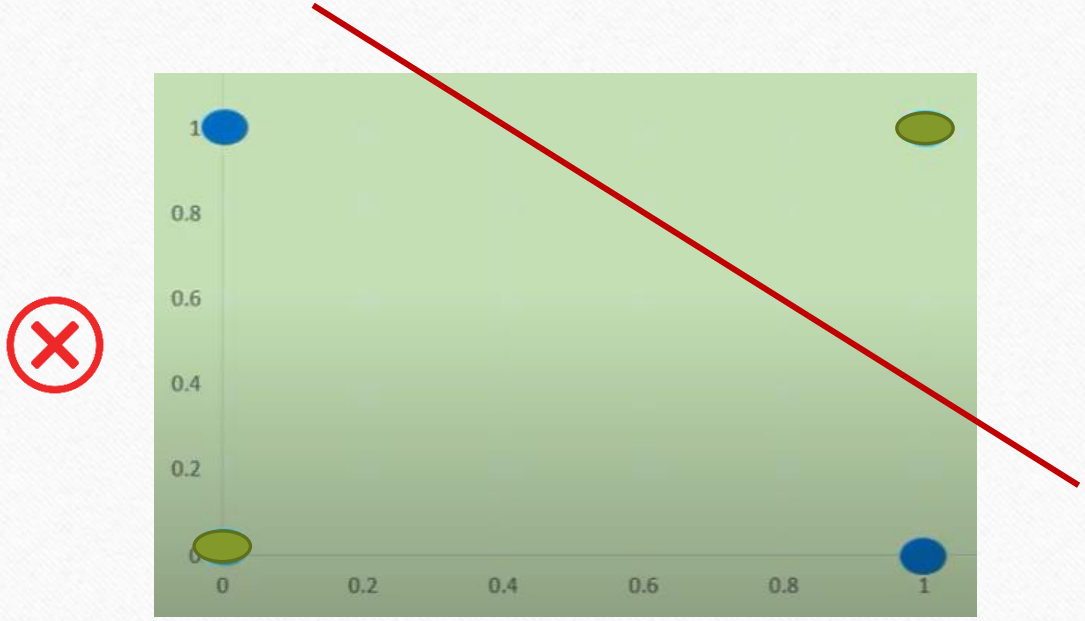
A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0

Exemple de classification: Résolution non linéaire



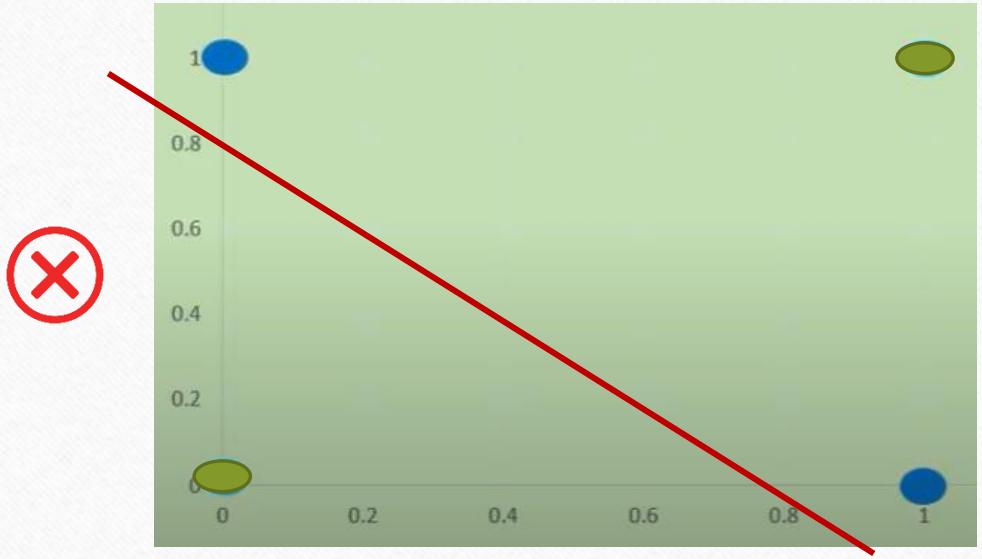
A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0

Exemple de classification: Résolution non linéaire



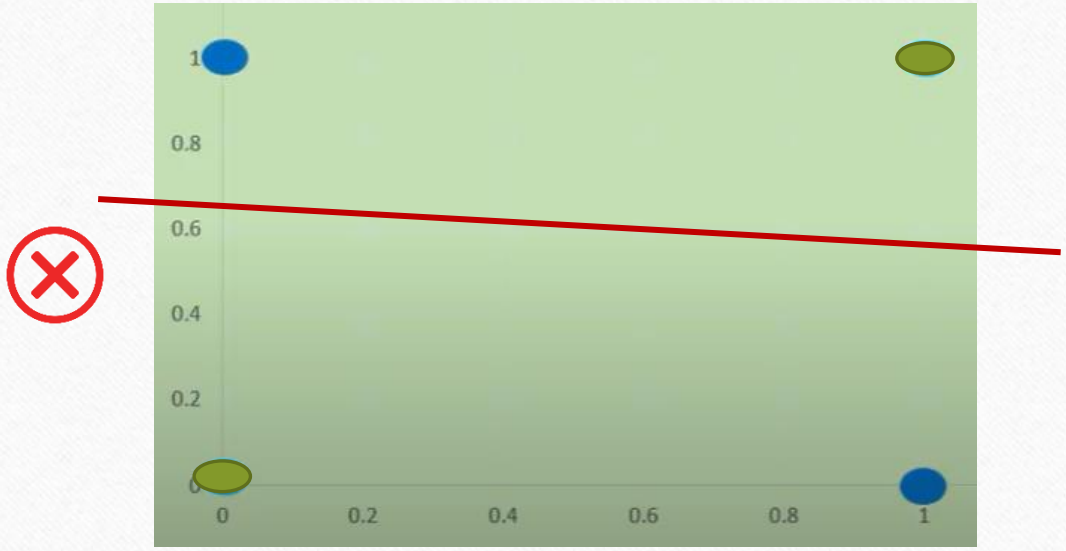
A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0

Exemple de classification: Résolution non linéaire



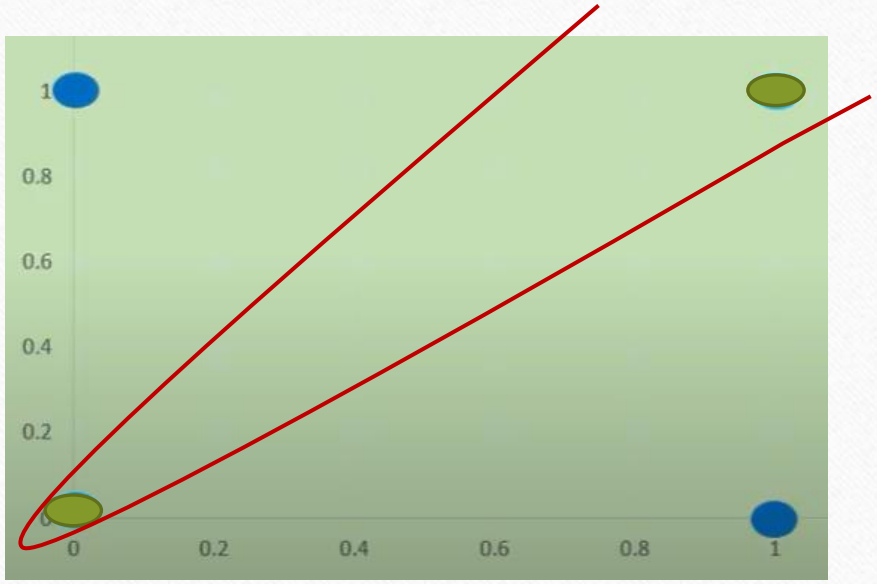
A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0

Exemple de classification: Résolution non linéaire



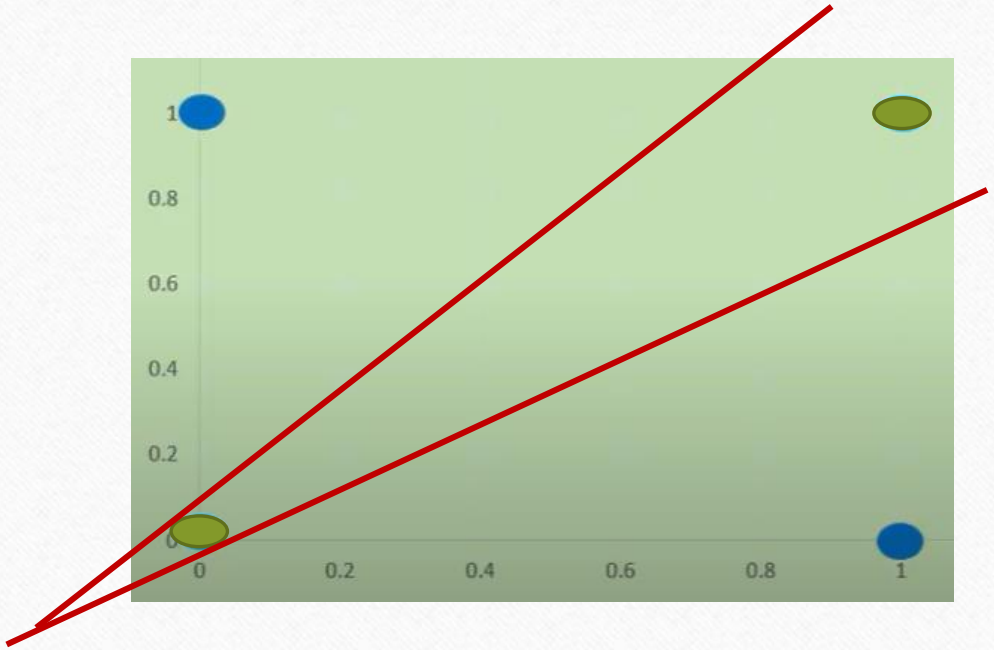
A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0

Exemple de classification: Résolution non linéaire

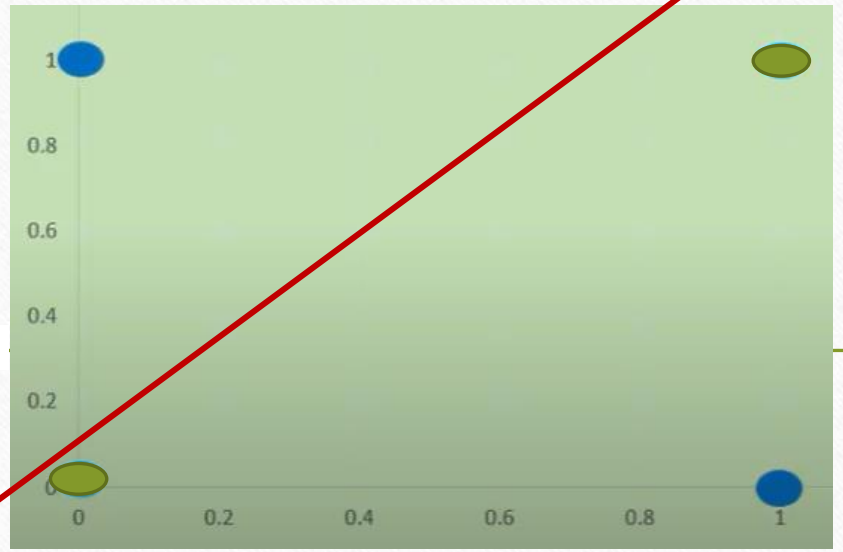


A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0

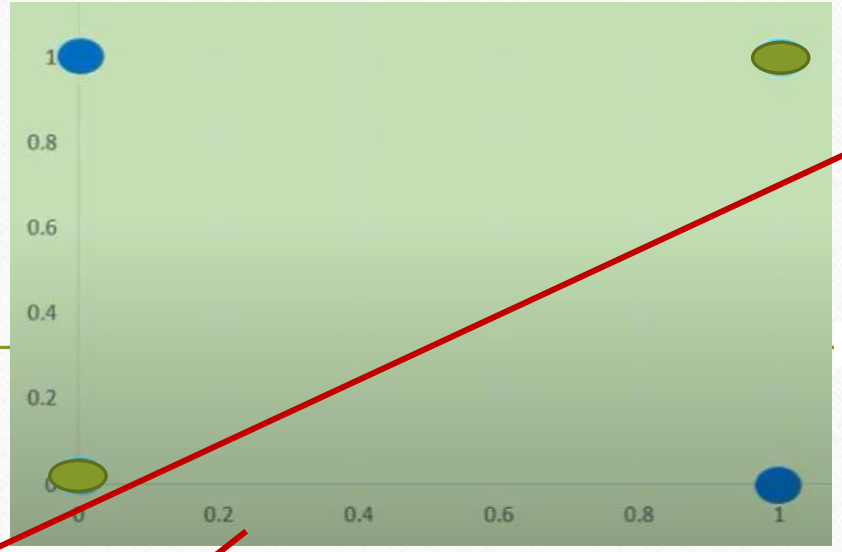
Exemple de classification



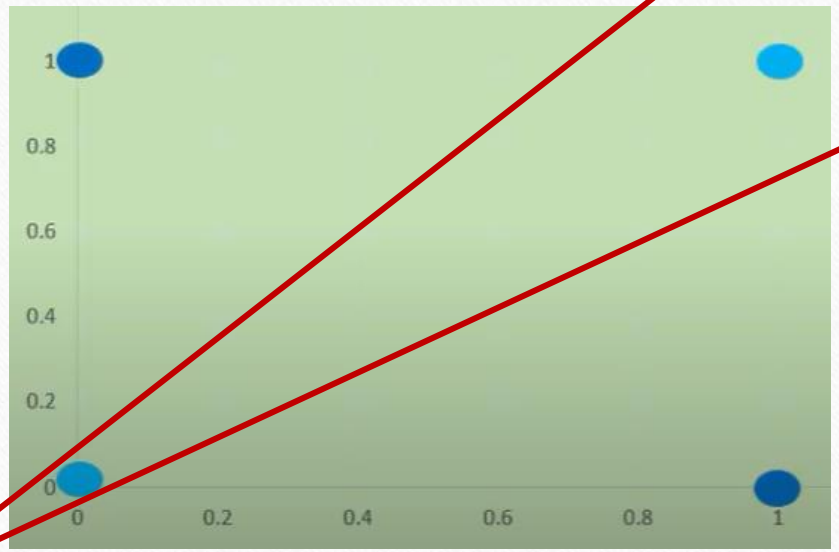
A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0


1
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1

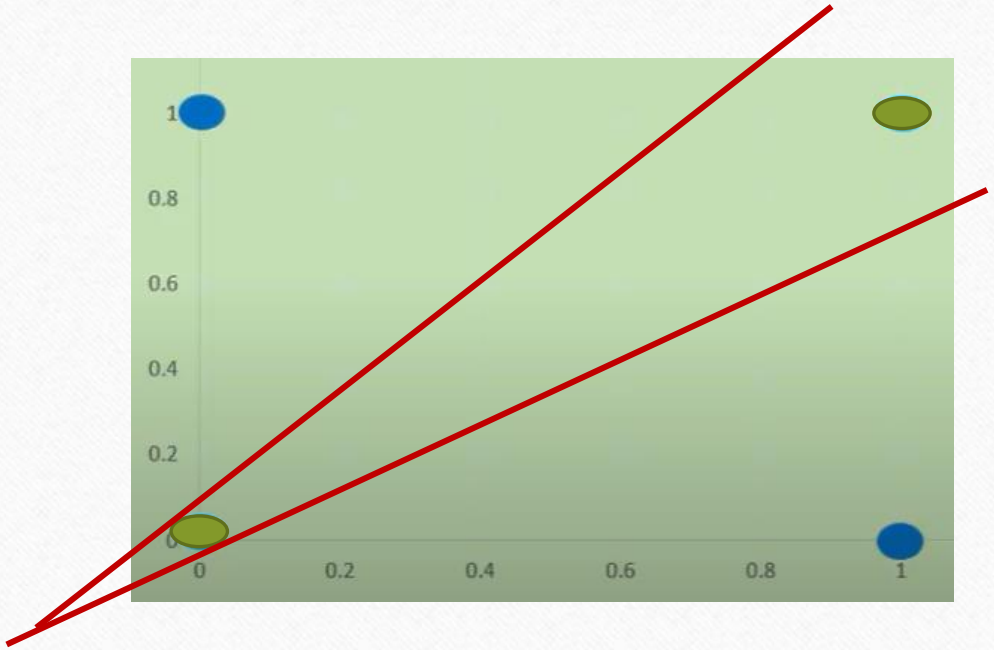
+


1
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1

=

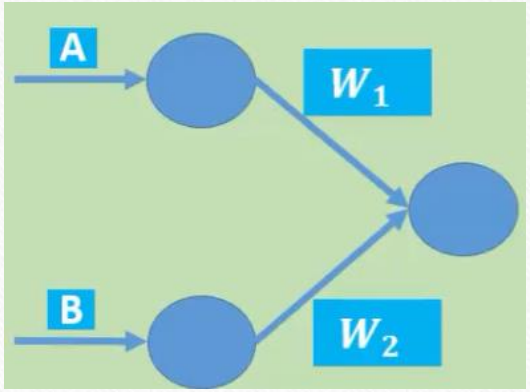

1
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1

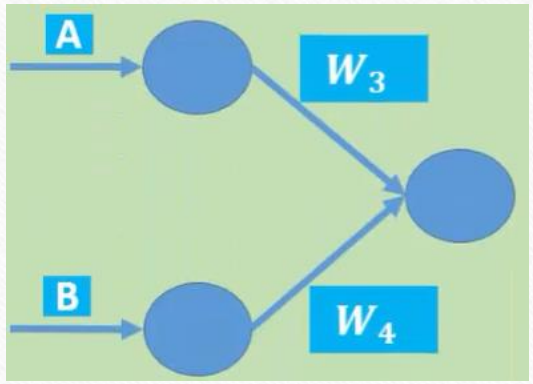
Exemple de classification



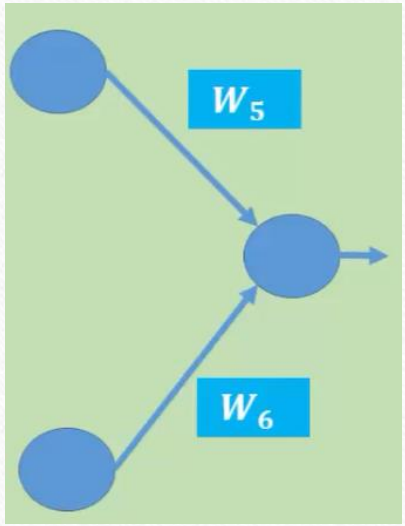
A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0

Exemple de classification: Architecture

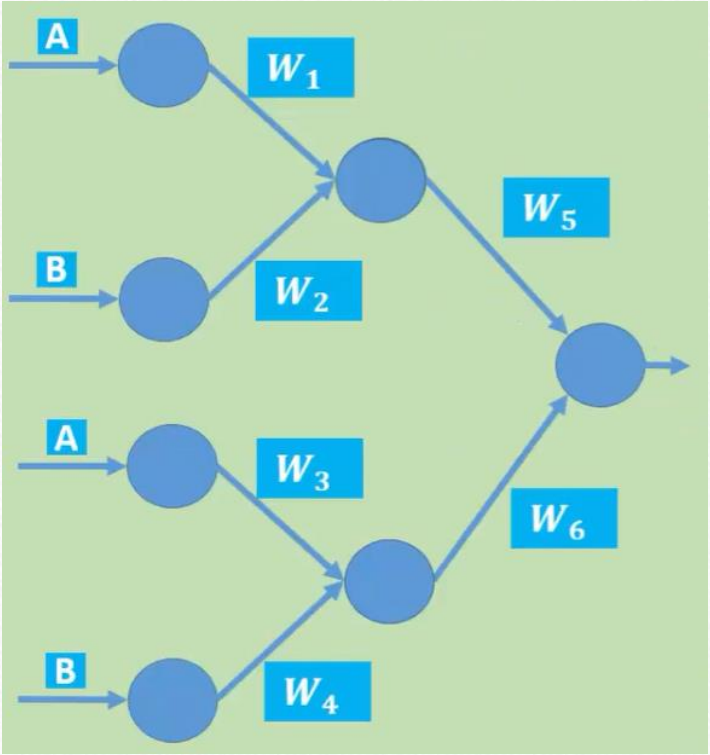


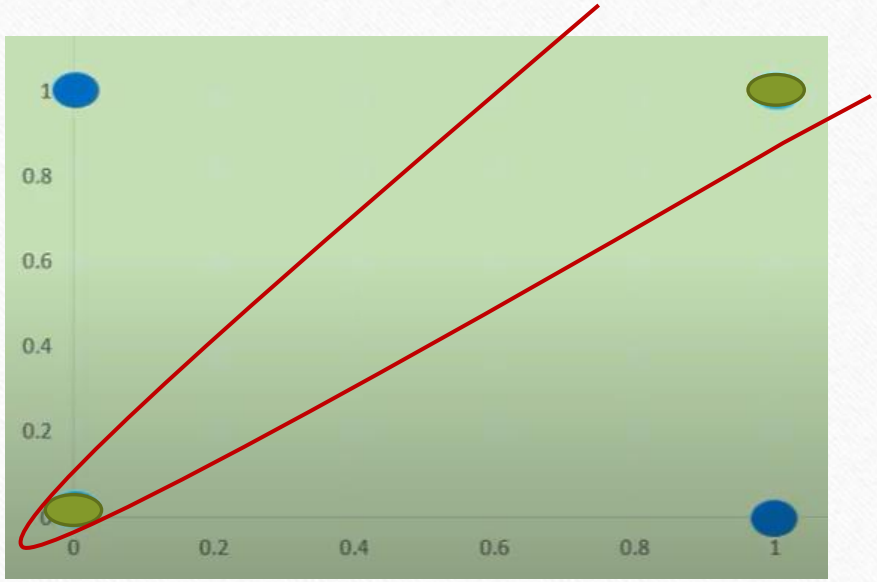


combinaison

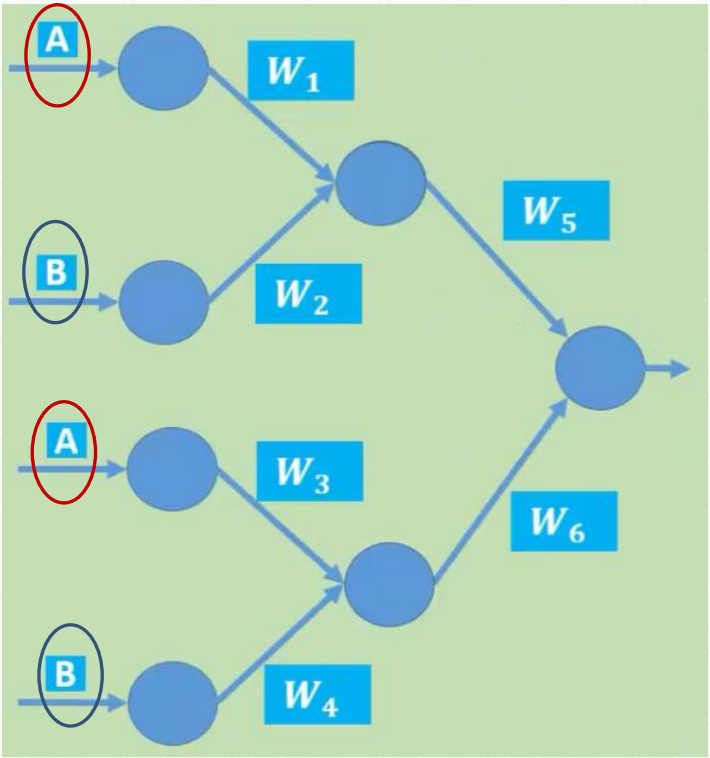


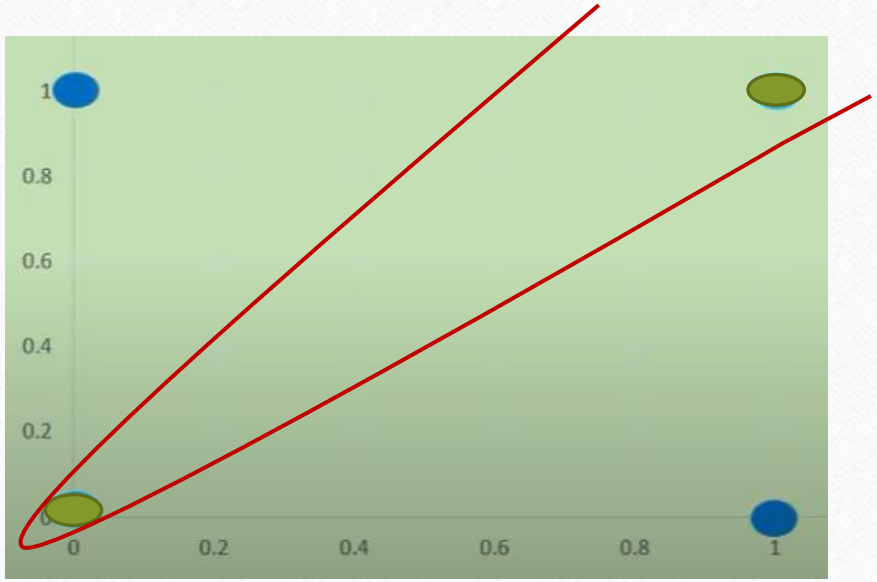
Exemple de classification: Architecture



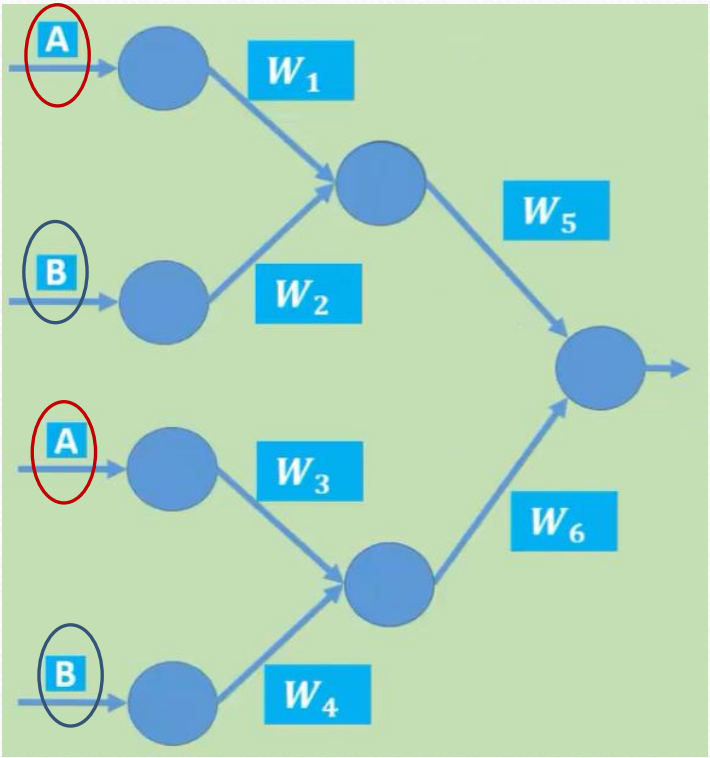


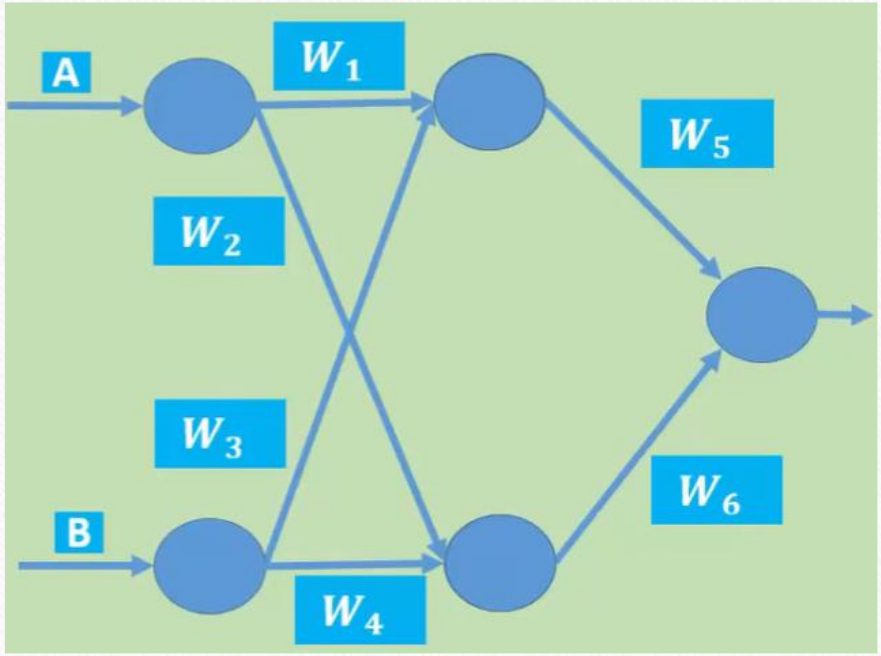
Exemple de classification: Architecture





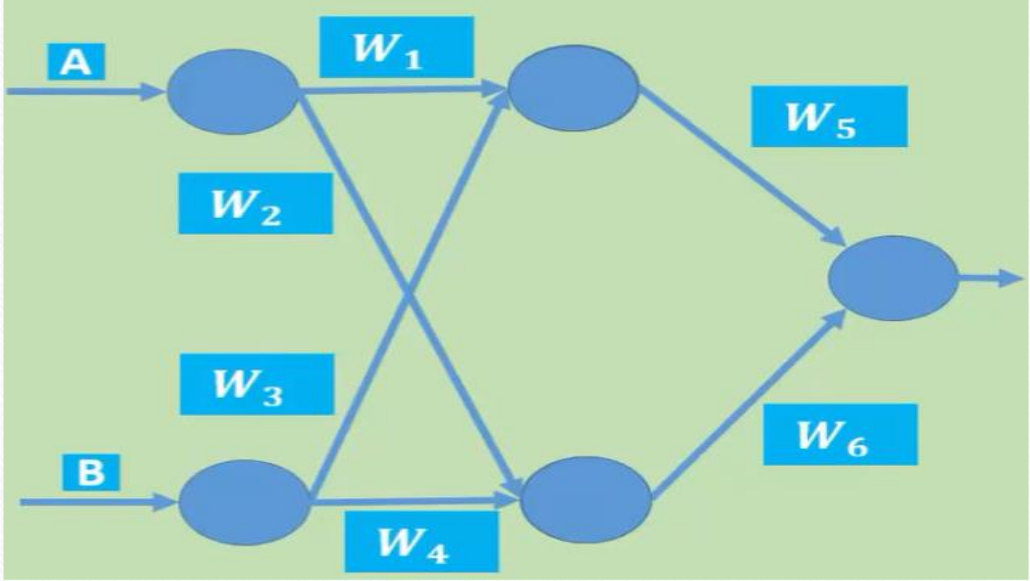
Exemple de classification: Architecture





Exemple de classification: Architecture

Couche d'entrée **Couche cachée** **Couche de sortie**



Les neurones d'une couche sont reliés à la totalité des neurones des couches adjacentes,

Etapes d'apprentissage

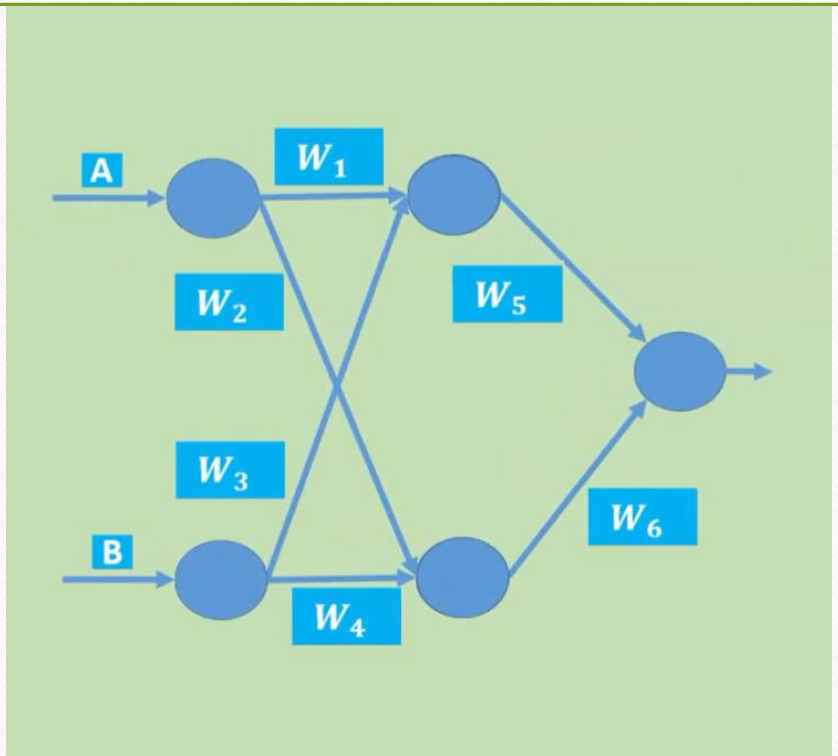
1. Initialiser les poids synaptiques
2. Passer les observations une à une
3. Calculer la somme des produits
4. Calculer la réponse de la fonction d'activation
5. Mise à jour des poids

Jusqu'à convergence

Exemple de classification: Apprentissage

A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0

- Pour chaque itération, les paramètres du réseau doivent être connus.



Exemple de classification: Apprentissage

- ***Paramètres à l'itération n = 0:***

η=0,001

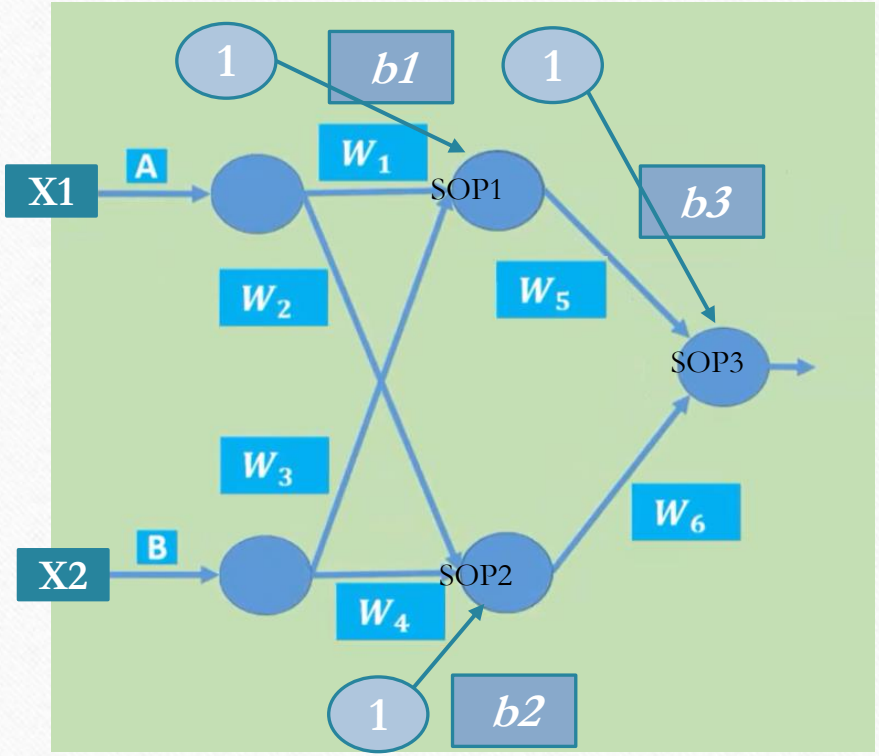
X(0)=[1,1,1,X_1,X_2]=[1,1,1, 1,0]

W(0) = [b1,b2,b3,W_1,W_2,W_3,W_4,W_5,W_6]

=[−1.5, −0.5, −0.5, 1,1,1,1,−2, 1]

d(0)=1

A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0



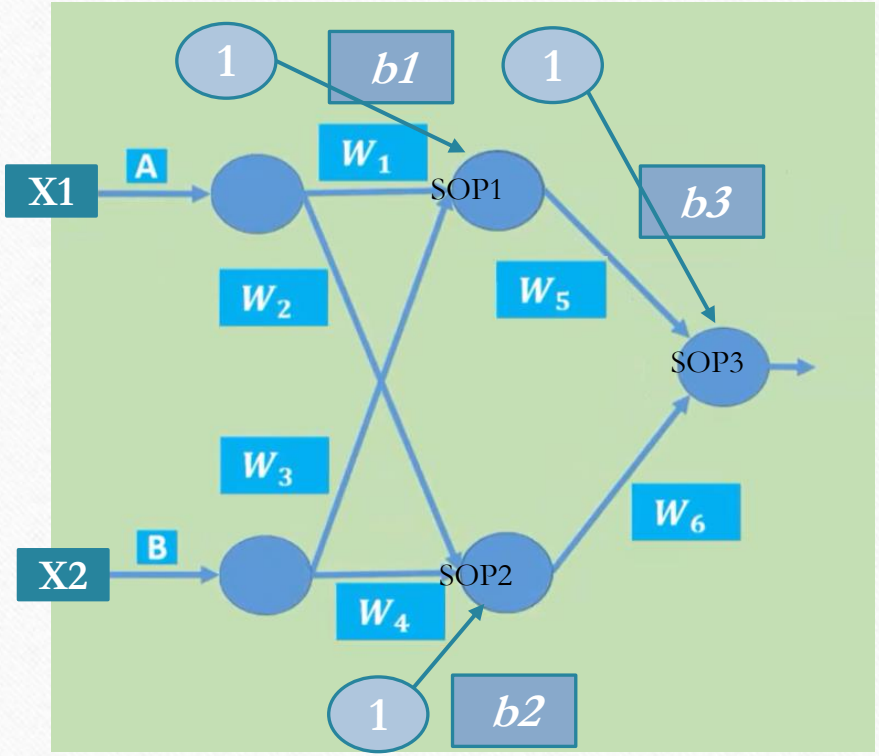
Exemple de classification: Apprentissage

- ***l'itération n = 0:***

SOP1 = (1*b1+X1*W1+X2*W3)
= -1.5 + 1 + 0 = **-0.5**

SOP2 = (1*b2+X1*W2+X2*W4)
= -0.5 + 1 + 0 = **0.5**

A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0



Exemple de classification: Apprentissage

- ***l'itération n = 0:***

SOP1 = (1*b1+X1*W1+X2*W3)
= -1.5 + 1 + 0 = **-0.5**

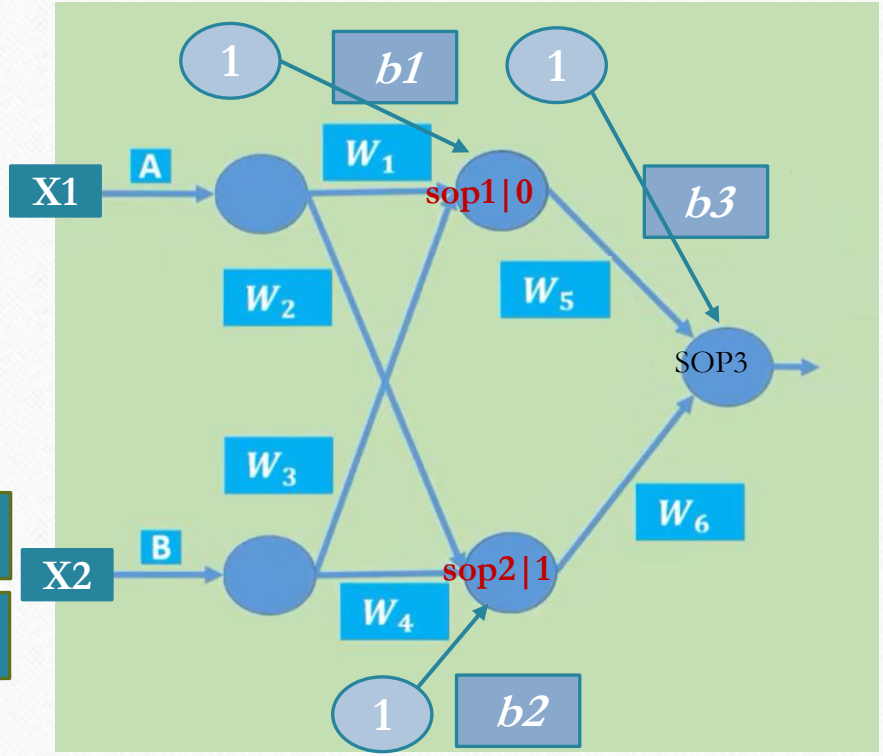
SOP2 = (1*b2+X1*W2+X2*W4)
= -0.5 + 1 + 0 = **0.5**

la **fonction de Heaviside**

$$H(x) = \begin{cases} 0 & \text{si} \quad x < 0 \\ 1 & \text{si} \quad x \geq 0. \end{cases}$$

Y1=Y(SOP1)=H(SOP1)=**0**

Y2=Y(SOP2)=H(SOP2)=**1**



Exemple de classification: Apprentissage

- ***l'itération n = 0:***

SOP1 = (1*b1+X1*W1+X2*W3)
= -1.5 + 1 + 0 = **-0.5**

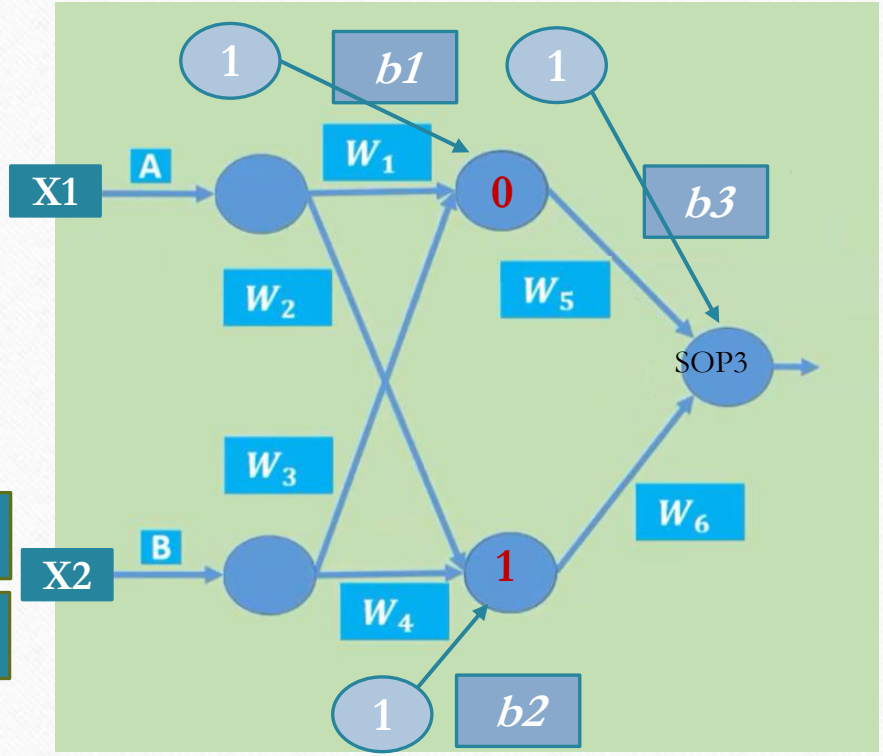
SOP2 = (1*b2+X1*W2+X2*W4)
= -0.5 + 1 + 0 = **0.5**

la **fonction de Heaviside**

$$H(x) = \begin{cases} 0 & \text{si} \quad x < 0 \\ 1 & \text{si} \quad x \geq 0. \end{cases}$$

Y1=Y(SOP1)=H(SOP1)=**0**

Y2=Y(SOP2)=H(SOP2)=**1**

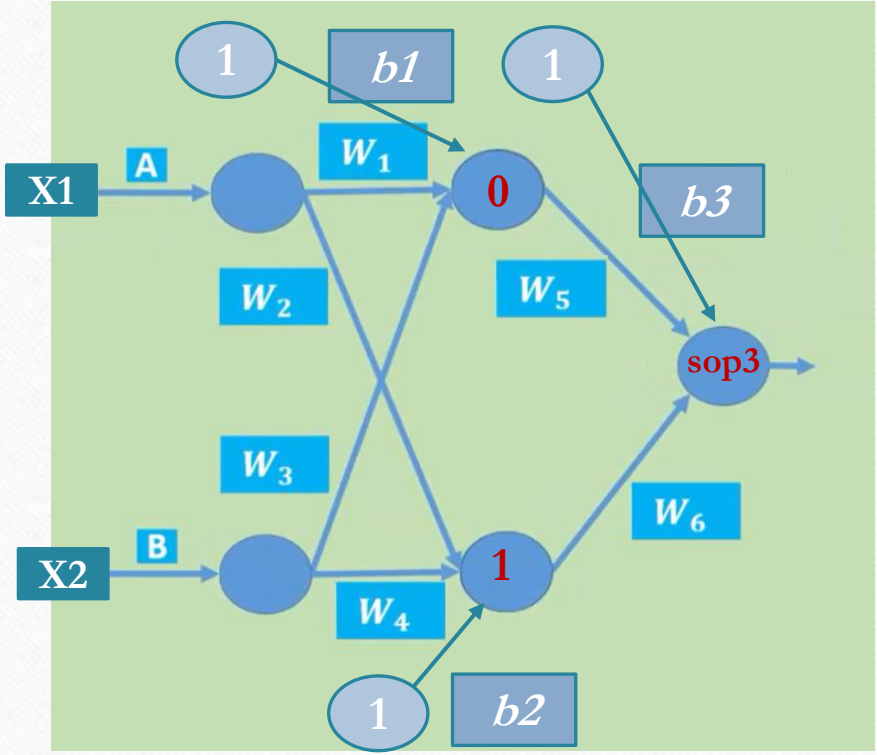


Exemple de classification: Apprentissage

- ***l'itération n = 0:***

SOP3 = (1*b3+X1*W5+X2*W6)
= (1*-0.5) + (0*-2) + (1*1) = **0.5**

Y = H(SOP3) = **1**



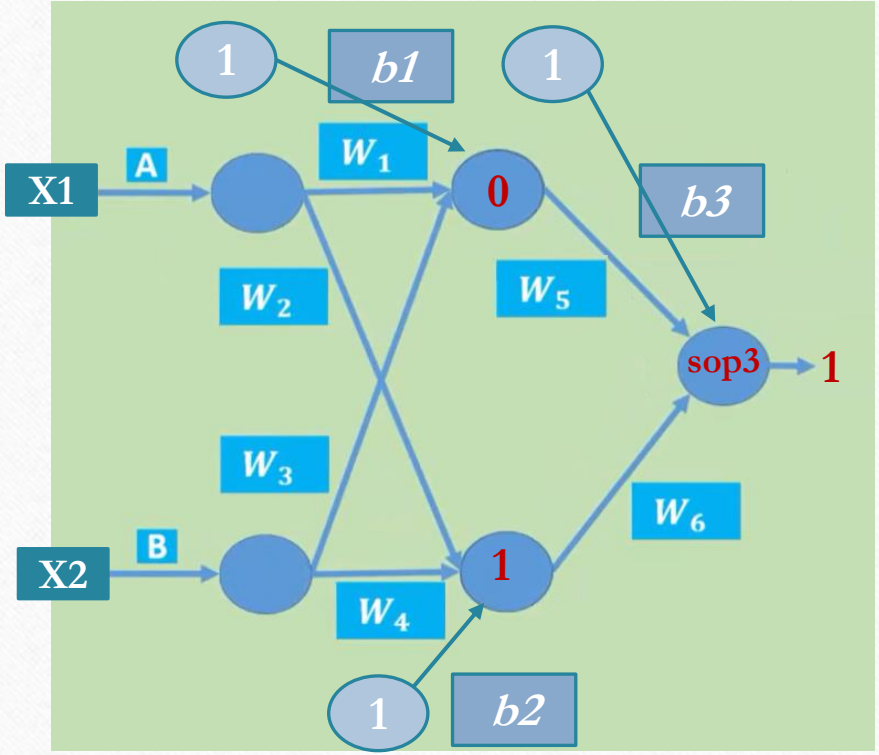
Exemple de classification: Apprentissage

- ***l'itération n = 0:***

SOP3 = (1*b3+X1*W5+X2*W6)
= (1*-0.5) + (0*-2) + (1*1) = **0.5**

Y = H(SOP3) = **1**

Y(0) = 1
d(0) = 1
Poids corrects
Pas de mise à jour



Exemple de classification: Apprentissage

- ***Paramètres à l'itération n = 1:***

η=0,001

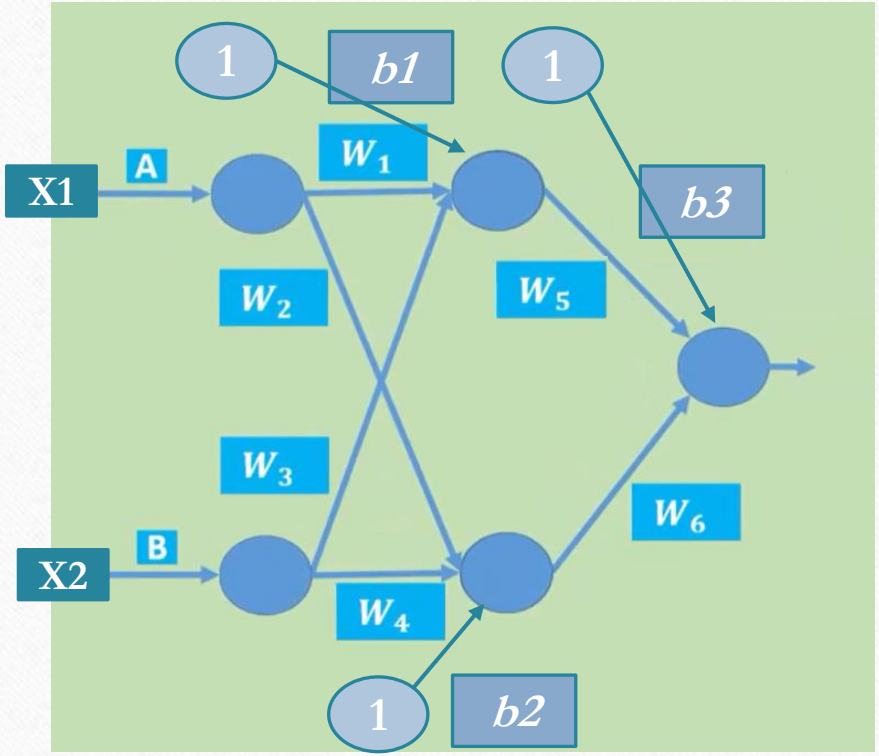
X(1)=[1,1,1,X_1,X_2]=[1,1,1, 0,1]

$W(1)$ = [b1,b2,b3,W_1,W_2,W_3,W_4,W_5,W_6]

=[−1.5, −0.5, −0.5, 1,1,1,1,−2, 1]

d(1)=1

A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0



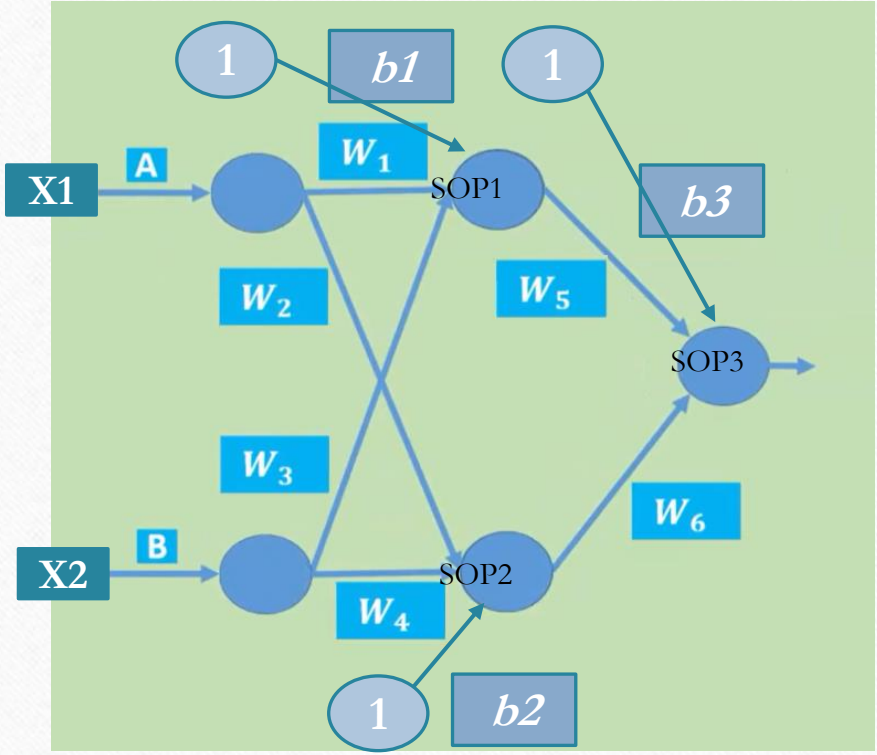
Exemple de classification: Apprentissage

- ***l'itération n = 1:***

SOP1 = (1*b1+X1*W1+X2*W3)
= -1.5 + 0 + 1 = **-0.5**

SOP2 = (1*b2+X1*W2+X2*W4)
= -0.5 + 0 + 1 = **0.5**

A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0



Exemple de classification: Apprentissage

- ***l'itération n = 1:***

SOP1 = (1*b1+X1*W1+X2*W3)
= -1.5 + 0 + 1 = **-0.5**

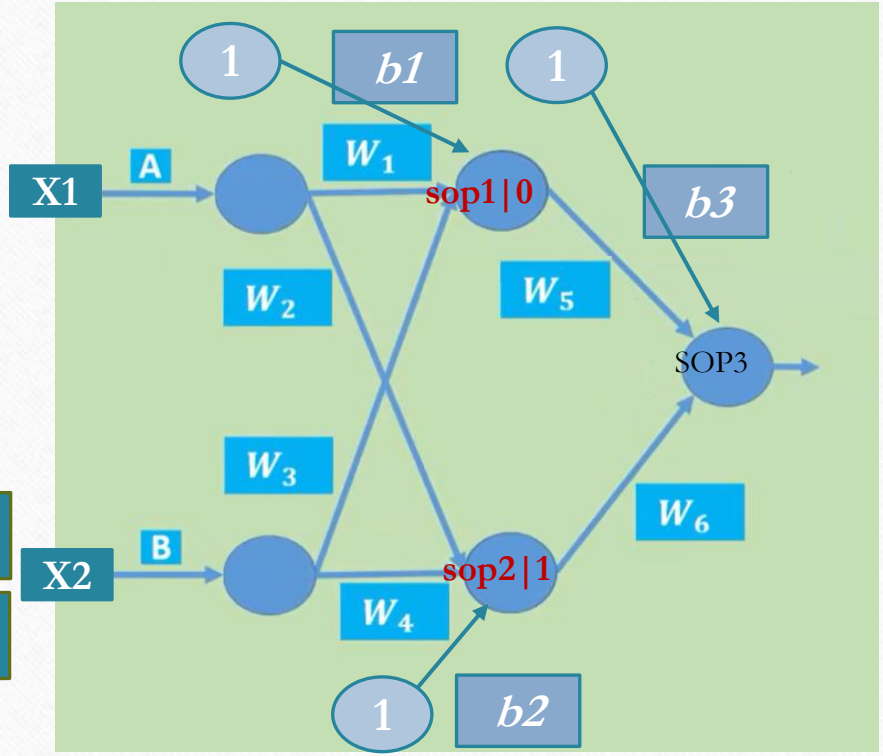
SOP2 = (1*b2+X1*W2+X2*W4)
= -0.5 + 0 + 1 = **0.5**

la **fonction de Heaviside**

$$H(x) = \begin{cases} 0 & \text{si} \quad x < 0 \\ 1 & \text{si} \quad x \geq 0. \end{cases}$$

Y1=Y(SOP1)=H(SOP1)=**0**

Y2=Y(SOP2)=H(SOP2)=**1**



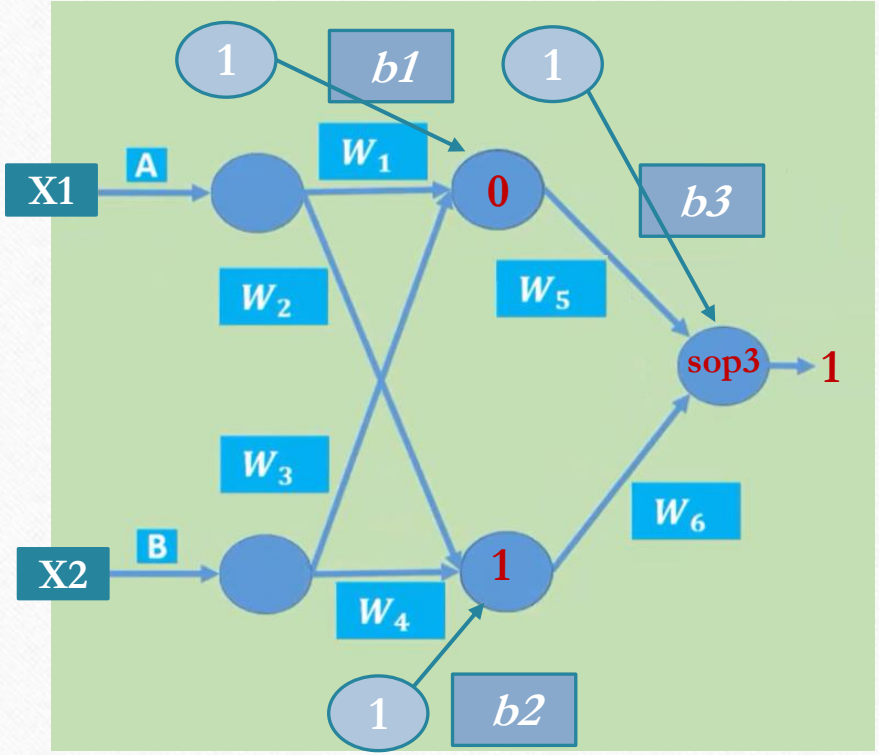
Exemple de classification: Apprentissage

- ***l'itération n = 1:***

SOP3 = (1*b3+X1*W5+X2*W6)
= (1*-0.5) + (0*-2) + (1*1) = **0.5**

Y = H(SOP3) = **1**

Y(1) = 1
d(1) = 1
Poids corrects
Pas de mise à jour



Exemple de classification: Apprentissage

- ***Paramètres à l'itération n = 2:***

η=0,001

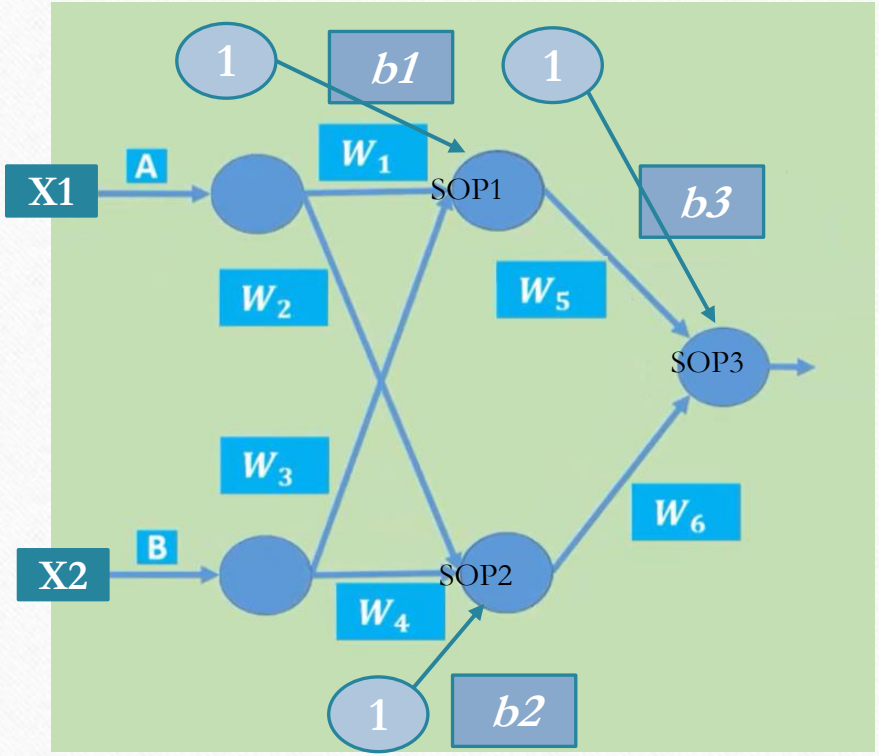
X(2)=[1,1,1,X_1,X_2]=[1,1,1, 0,0]

W(2) = [b1,b2,b3,W_1,W_2,W_3,W_4,W_5,W_6]

=[−1.5, −0.5, −0.5, 1,1,1,1,−2, 1]

d(2)=0

A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0



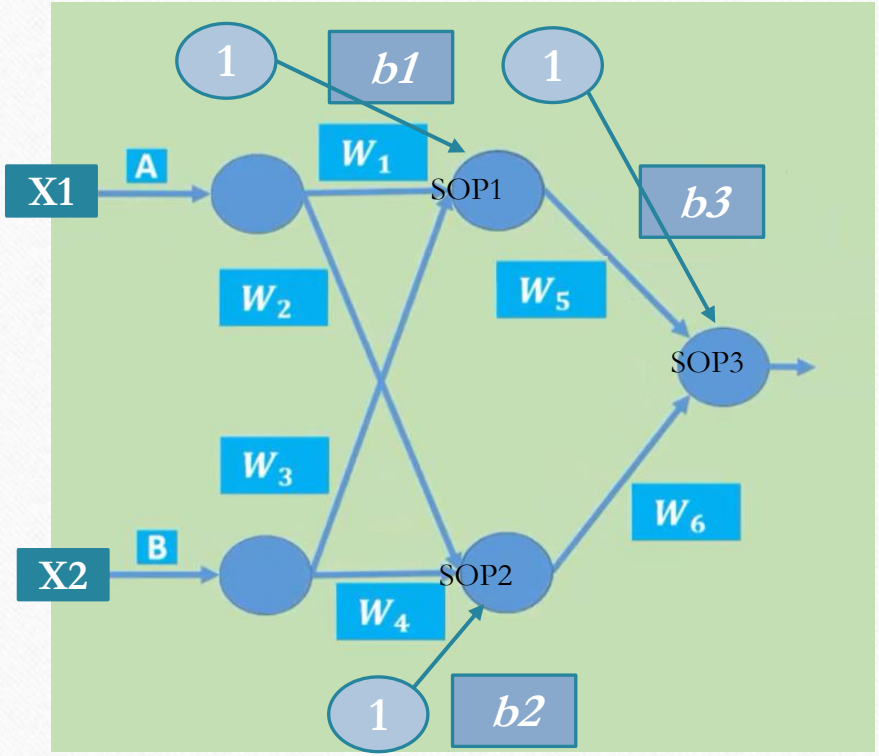
Exemple de classification: Apprentissage

- ***l'itération n = 2:***

SOP1 = (1*b1+X1*W1+X2*W3)
= -1.5 + 0 + 0 = **-1.5**

SOP2 = (1*b2+X1*W2+X2*W4)
= -0.5 + 0 + 0 = **-0.5**

A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0



Exemple de classification: Apprentissage

- ***l'itération n = 2:***

SOP1 = (1*b1+X1*W1+X2*W3)
= -1.5 + 0 + 0 = **-1.5**

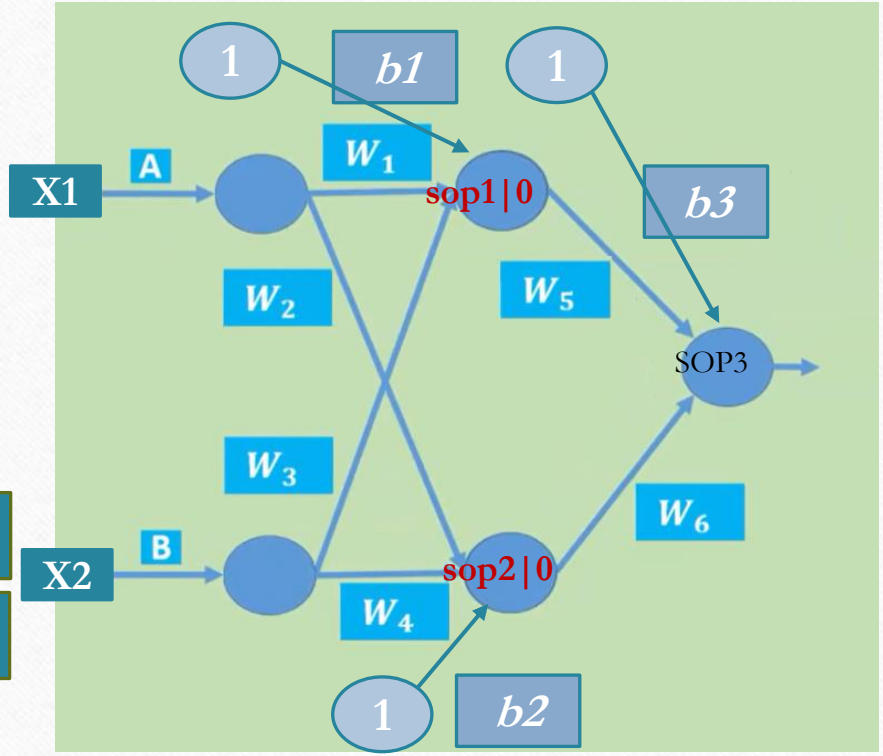
SOP2 = (1*b2+X1*W2+X2*W4)
= -0.5 + 0 + 0 = **-0.5**

la **fonction de Heaviside**

$$H(x) = \begin{cases} 0 & \text{si} \quad x < 0 \\ 1 & \text{si} \quad x \geq 0. \end{cases}$$

Y1=Y(SOP1)=H(SOP1)=**0**

Y2=Y(SOP2)=H(SOP2)=**0**



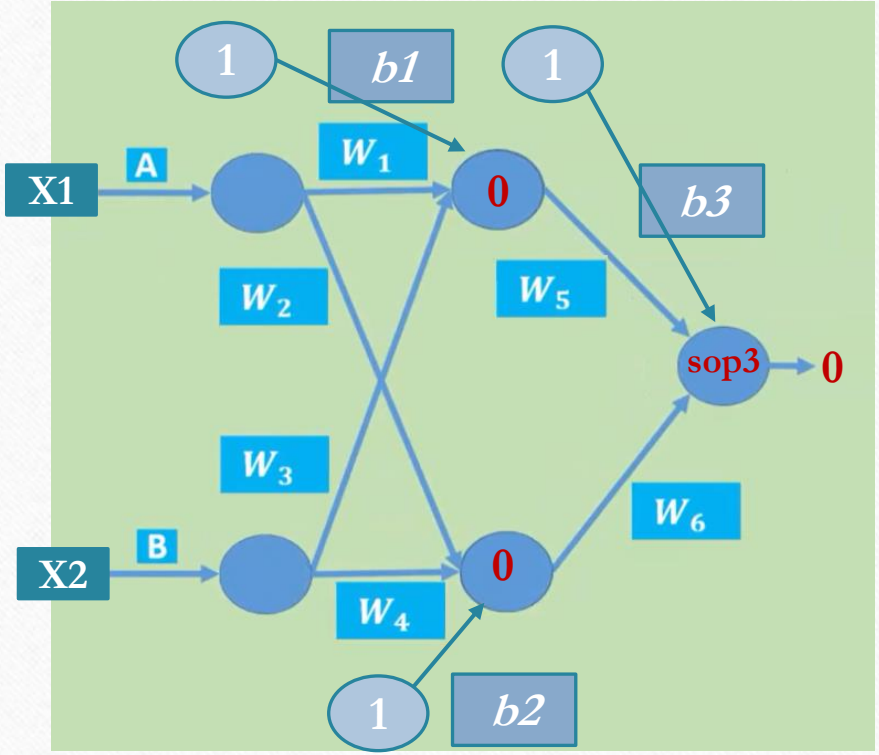
Exemple de classification: Apprentissage

- ***l'itération n = 2:***

SOP3 = (1*b3+X1*W5+X2*W6)
= (1*-0.5) + (0*-2) + (0*1) = **-0.5**

Y = H(SOP3) = **0**

Y(2) = 0
d(2) = 0
Poids corrects
Pas de mise à jour



Exemple de classification: Apprentissage

- ***Paramètres à l'itération n = 3:***

η=0,001

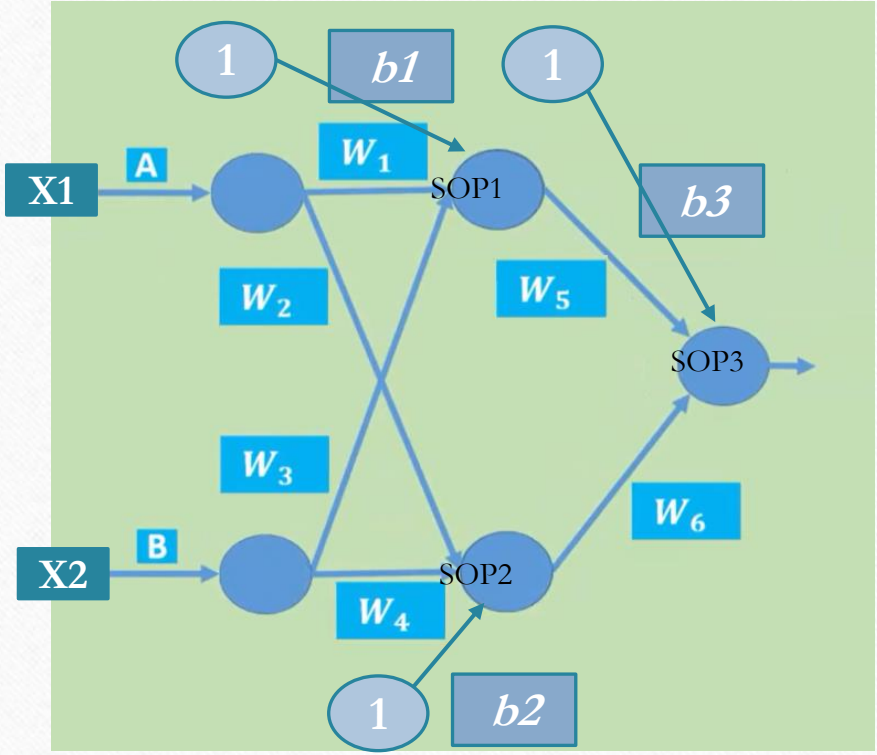
X(3)=[1,1,1,X_1,X_2]=[1,1,1, 1,1]

W(3) = [b1,b2,b3,W_1,W_2,W_3,W_4,W_5,W_6]

=[−1.5, −0.5, −0.5, 1,1,1,1,−2, 1]

d(3)=0

A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0



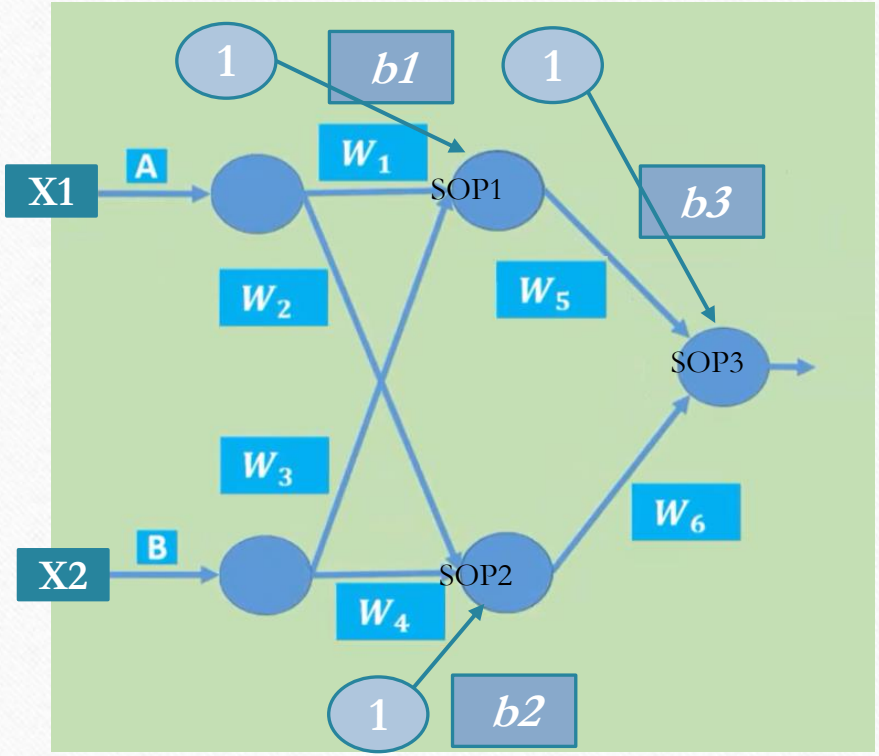
Exemple de classification: Apprentissage

- ***l'itération n = 3:***

SOP1 = (1*b1+X1*W1+X2*W3)
= -1.5 + 1 + 1 = **0.5**

SOP2 = (1*b2+X1*W2+X2*W4)
= -0.5 + 1 + 1 = **1.5**

A	B	Classe
1	0	1
0	1	1
0	0	0
1	1	0



Exemple de classification: Apprentissage

- ***l'itération n = 3:***

SOP1 = (1*b1+X1*W1+X2*W3)
= -1.5 + 1 + 1 = **0.5**

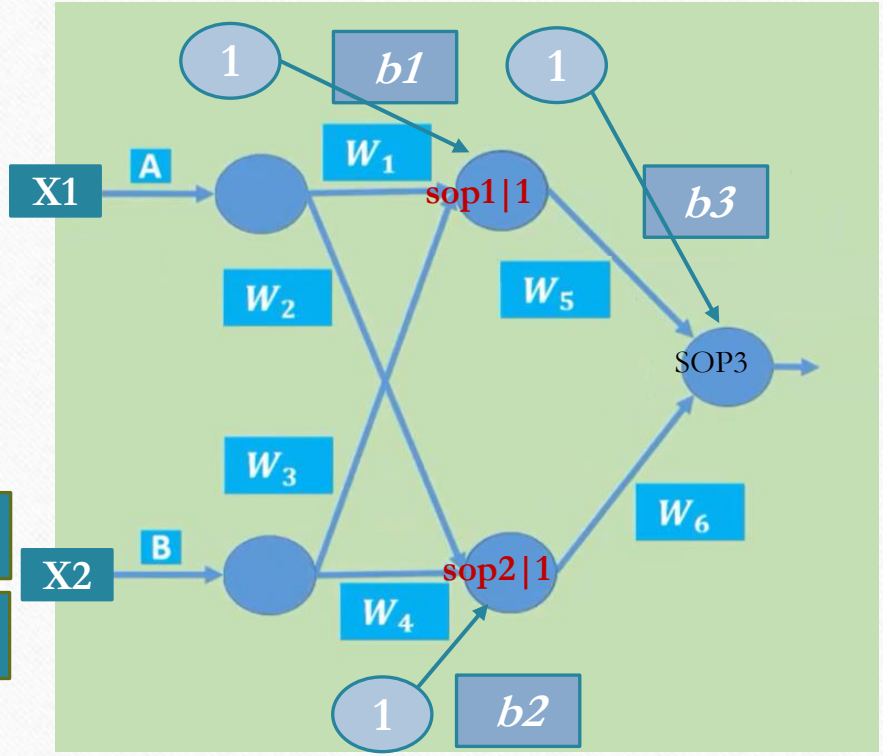
SOP2 = (1*b2+X1*W2+X2*W4)
= -0.5 + 1 + 1 = **1.5**

la **fonction de Heaviside**

$$H(x) = \begin{cases} 0 & \text{si} \quad x < 0 \\ 1 & \text{si} \quad x \geq 0. \end{cases}$$

Y1=Y(SOP1)=H(SOP1)=**1**

Y2=Y(SOP2)=H(SOP2)=**1**



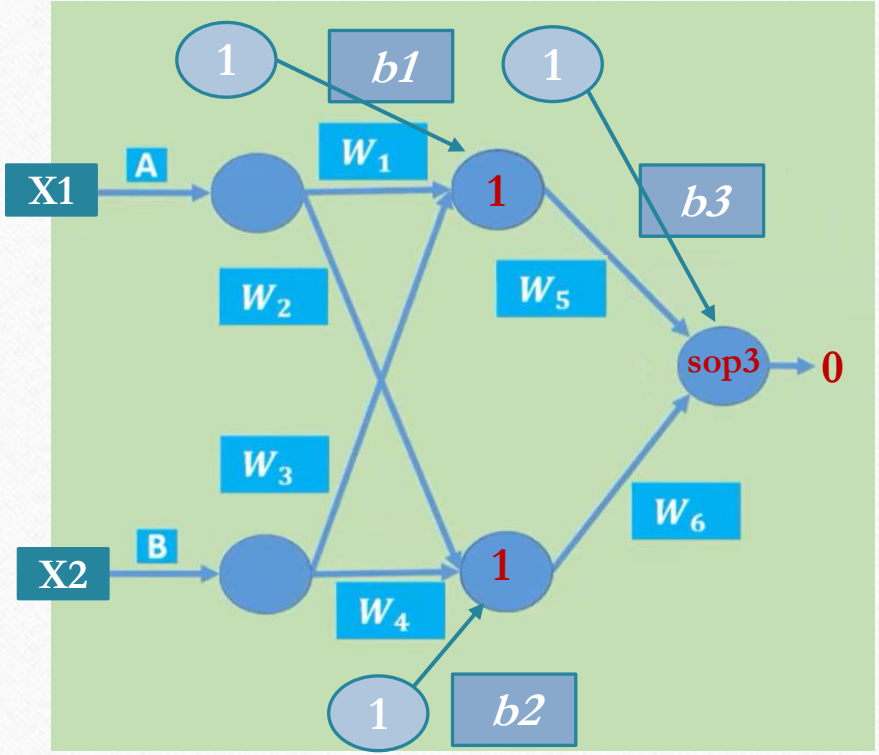
Exemple de classification: Apprentissage

- ***l'itération n = 3:***

SOP3 = (1*b3+X1*W5+X2*W6)
= (1*-0.5) + (1*-2) + (1*1) = **-1.5**

Y = H(SOP3) = **0**

Y(3) = 0
d(3) = 0
Poids corrects
Pas de mise à jour



Exemple de classification: Apprentissage

- Les poids considérés par ce modèle sont:

$$\mathbf{W=[-1.5, -0.5, -0.5, 1,1,1,1,-2, 1]}$$